# McGill University
# Department of Mathematics and Statistics
# MATH 782 Advanced Topics in Statistics

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Degree Program and Year: PhD 2  Term: Fall  Winter

Topics to be covered: (Outline the Main reading material.)

Topic Title: Asymptotic Statistics

The course aims to provide an introduction to modern tools and topics in asymptotic statistics. It will be split into three main parts: (A) mathematical and probabilistic foundations; (B) classical large-sample theory; (C) an introduction to minimax rates for nonparametric estimation and semiparametric theory.

In part (A), the focus will be on mathematical reminders, stochastic convergence, uniform integrability, laws of large numbers, central limit theorems, triangular arrays, and asymptotic linearity.
Part (B) will focus on studying the asymptotic properties of various classical estimators, such as the empirical distribution function and sample quantiles, maximum likelihood, M-estimation, Z-estimation, with various applications to estimation and testing.
Part (C) will be an introduction to selected modern topics such as minimax upper/lower bounds for nonparametric regression and semiparametric theory, as time permits.

**Method of Evaluation:** Assignments: 25%. Project: 50 %. Oral presentation: 25%.

Student Signature and Date:

Instructor Name: Mehdi Dagdoug  Signature and Date: Dagdoug

**For Undergraduate Students - Chief Advisor Approval**:

Professor Djivede Kelome  Signature and Date:

**For Graduate Students – Graduate Program Director Approval**

Professor Brent Pym  Signature and Date:

**Please return form with all 3 signature via email to the Undergraduate Program Coordinator, Angela White at angela.white@mcgill.ca**

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